

Cognitive Load

Cognitive task analysis and workload classification

Table 1
Bloom's Taxonomy listed in order of increasing cognitive complexity.

	Taxonomy Level	Description	Example Verbs
1	Knowledge	Recall of specific facts or ideas.	remember, define, list, memorize
2	Comprehension	Understanding and interpreting facts and ideas.	classify, explain, discuss, identify
3	Application	The use of prior knowledge in novel situations.	execute, solve, operate, respond
4	Analysis	Decomposing a system into its composite parts and examining those parts.	compare, associate, contrast, test
5	Synthesis	Combining independent elements to form a new system.	assemble, design, integrate, produce
6	Evaluation	Judging the value of a system based on evidence and certain criteria.	judge, appraise, defend, critique

Une tache peut appartenir à un niveau de taxonomie ?

Table 2
Harrow's Taxonomy listed in order of increasing psychomotor complexity.

	Taxonomy Level	Description	Example Verbs
1	Reflexive Movements	Involuntary movements evoked in response to some stimuli.	flex, extend, stretch, react
2	Fundamental Movements	Basic movement patterns which build on reflexive movements.	reach, grasp, walk, jump, crawl
3	Perceptual Abilities	Ability to receive information about oneself and the world via one of several sensory systems (vision, hearing, etc.).	sense, perceive, hear, see, feel
4	Physical Abilities	The functional characteristics of the body which govern the efficiency of skills in the psychomotor domain.	exert endurance, exert strength, exert flexibility
5	Skilled Movements	Complex movement skills which require learning.	dance, drive, juggle
6	Non-Discursive Communication	Learned movements and gestures used for communication.	express, posture, gesture

The Significance of Human Performance in Production Processes: An Extensive Review of Simulation-Integrated Techniques for Assessing Fatigue and Workload

Table 1. VACP Values and Descriptors [19]

Scale	Descriptor	Scale	Descriptor	Scale	Descriptor	Scale	Descriptor
Visual		Auditory		Cognitive		Psychomotor	
0.0	No visual activity	0.0	No auditory activity	0.0	No cognitive activity	0.0	No Psychomotor activity
1.0	Visually detect	1.0	Detect sound	1.0	Automatic	1.0	Speech
3.7	Visually discriminate	2.0	Orient to sound(general)	1.2	Alternative selection	2.2	Discrete actuation
4.0	Visually check	4.2	Orient to sound(selective)	3.7	Signal recognition	2.6	Continuous adjustive
5.0	Visually locate	4.3	Verify feedback	4.6	Evaluation (single aspect)	4.6	Manipulative
5.4	Visually track	4.9	Interpret speech	5.3	Encoding/Decodin g	5.8	Discrete adjustive
5.9	Visually read	6.6	Discriminate sound	6.8	Evaluation (several aspect)	6.5	Symbolic production
7.0	Visually search	7.0	Interpret sound patterns	7.0	Estimation, calculation, conversion	7.0	Serial Discrete Manipulation

Table 2. Rest allowance and Fatigue models

Rohmert (1973)	$RA = 18 \times (fMHT)^{1.4} \times fMVC - 0.15)^{0.5} \times 100$	RA state for rest allowances fMVC is the fraction of maximum voluntary contraction of a muscle to perform any task.and fMHT is fraction of maximum holding time
Rose et al.(1992)	$RA = 0.164x \left[4.61 + \ln \left(\frac{1}{100 - (fMHT)^{-1}} \right) \right]^{-1} \times 100$ $MET = (7.96)e^{-4.16(\%MVC)}$	fMHT required
Ma et al. (2009)	$\frac{F_{cem}(t)}{MVC} = e^{\int_0^t -[F_{load}(u)/MVC]du}$	$F_{cem}(t)$ is current exertable maximum force (current muscle strength); $F_{load}(t)$ = forces required for the task (e.g., workloads); and t = current time (seconds)
Jaber and Neumann (2010)	$L_{max}^i = MLC * f_i * MET_i$	L_{max}^i is the maximum fatigue index for task i; MLC is maximum load capability; f_i is an individual's maximum capability
Perez et al. (2014)	$fF_i = \frac{t_i}{MET_i}$	fF_i is the fraction of fatigue contribution per task i; MET_i is the maximum endurance time for task i.
Dode et al. (2016)	$MET = (7.96)e^{-4.16(\%MVC/100)}$ $RA = 3MET^{-1.52}$	MVC is the maximum voluntary contraction of a muscle to perform any task.
Glock et al. (2019)	$F_s(t_a) = 1 - e^{-\theta_a t_a}$ $F_r(t_w) = \beta F_s(t_w)$	$F_s(t_a)$ is the fatigue accumulated by time t_a ; θ_a is a fatigue growth parameter when a worker completes an action a during time t_a ; $F_r(t_w)$ is the dynamic (real) fatigue.

Koruca, Halil İbrahim, Kemal Burak Urgancı, et Samia Chehbi Gamoura. « **The Significance of Human Performance in Production Processes: An Extensive Review of Simulation-Integrated Techniques for Assessing Fatigue and Workload** ». In *Advances in Intelligent Manufacturing and Service System Informatics*, édité par Zekâi Şen, Özer Uygun, et Caner Erden. Lecture Notes in Mechanical Engineering. Springer Nature Singapore, 2024. https://doi.org/10.1007/978-981-99-6062-0_51. **2**

HUMAN PERFORMANCE MODELING FOR DISCRETE-EVENT SIMULATION: WORKLOAD

Table 1: VACP Values and Descriptors

Scale Value	Scale Descriptor
Visual	
0.0	No Visual Activity
1.0	Visually Register/Detect (detect occurrence of image)
3.7	Visually Discriminate (detect visual differences)
4.0	Visually Inspect/Check (discrete inspection/static condition)
5.0	Visually Locate/Align (selective orientation)
5.4	Visually Track/Follow (maintain orientation)
5.9	Visually Read (symbol)
7.0	Visually Scan/Search/Monitor (continuous/serial inspection, multiple conditions)
Auditory	
0.0	No Auditory Activity
1.0	Detect/Register Sound (detect occurrence of sound)
2.0	Orient to Sound (general orientation/attention)
4.2	Orient to Sound (selective orientation/attention)
4.3	Verify Auditory Feedback (detect occurrence of anticipated sound)
4.9	Interpret Semantic Content (speech)
6.6	Discriminate Sound Characteristics (detect auditory differences)
7.0	Interpret Sound Patterns (pulse rates, etc.)
Cognitive	
0.0	No Cognitive Activity
1.0	Automatic (simple association)
1.2	Alternative Selection
3.7	Sign/Signal Recognition
4.6	Evaluation/Judgment (consider single aspect)
5.3	Encoding/Decoding, Recall
6.8	Evaluation/Judgment (consider several aspects)
7.0	Estimation, Calculation, Conversion
Psychomotor	
0.0	No Psychomotor Activity
1.0	Speech
2.2	Discrete Actuation (button, toggle, trigger)
2.6	Continuous Adjustive (flight control, sensor control)
4.6	Manipulative
5.8	Discrete Adjustive (rotary, vertical thumbwheel, lever position)
6.5	Symbolic Production (writing)
7.0	Serial Discrete Manipulation (keyboard entries)

Table 3: Task VACP Workload Values

Task	Vis.	Aud.	Cog.	Psy.
Driving	6	1	3.7	2.6
Stopped at Light	3	1	3.7	0
Start after stop	6	1	4.6	2.6
Dial and Press Send	5	4.3	5.3	7
Wait to connect	0	4.3	3.7	2.6
Talk	0	6	6.8	2.6

Cognitive Load

Pour 2 tâches fait en simultané on additionne les workloads de chaque catégorie.

Pour le workload total on additionne les workloads des catégories.

3 cite le 4 pour son tableau VACP

Lorsque la tâche est terminée les workloads associé chutent à 0

Keller, John. "Human performance modeling for discrete-event simulation: workload." *Proceedings of the Winter Simulation Conference*. Vol. 1. IEEE, 2002. - cité 82 fois

Aldrich, T. B., Szabo, S. M., & Bierbaum, C. R. (1989). The development and application of models to predict operator workload during system design. In G. R. McMillan, D. Beevis, E. Salas, M. H. Strub, R. Sutton, & L. Van Breda (Eds.), *Applications of human performance models to system design* (pp. 65–80). New York and London: Plenum. - cité 143 fois

Prediction of Mental Workload in Single and Multiple Tasks Environments

5 cite le VACP du 4

Pour prédire la charge mentale, le modele VACP et d'autre (TLAP, W/INDEX) prennent en compte le fait que des taches peuvent être fait en parallèle

Prédire la charge mentale en prenant en considération des caractéristiques individuel

Modélisation de la charge mentale en 2 parties :

- **Effective workload** et **ineffective workload**

Propose plusieurs type de mesure de workload

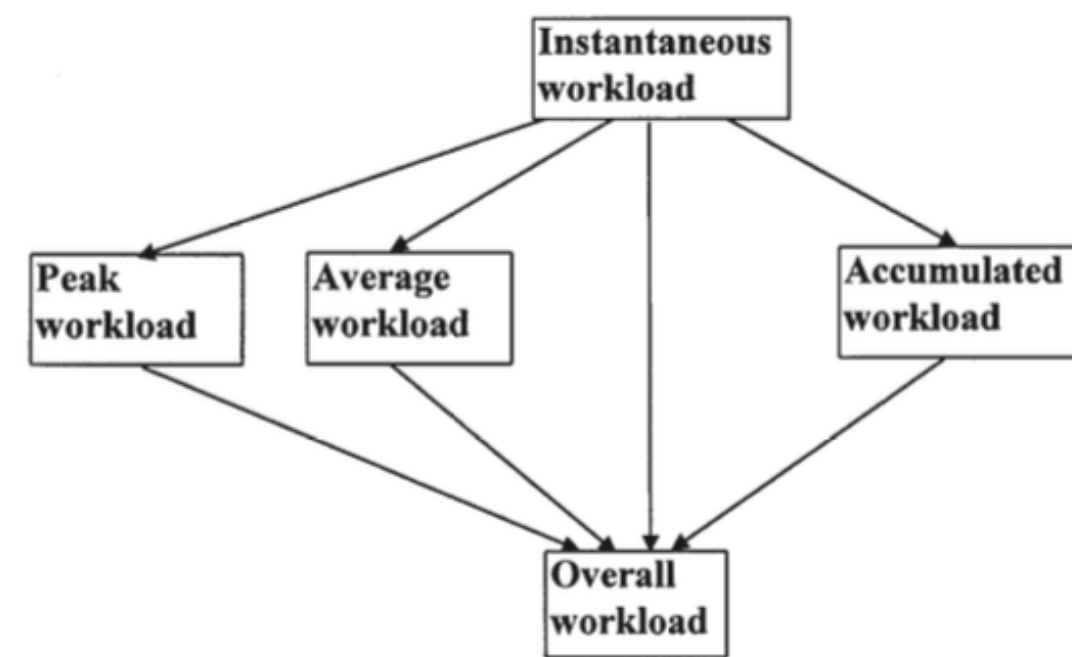


FIGURE 1 Conceptual framework of mental workload types.

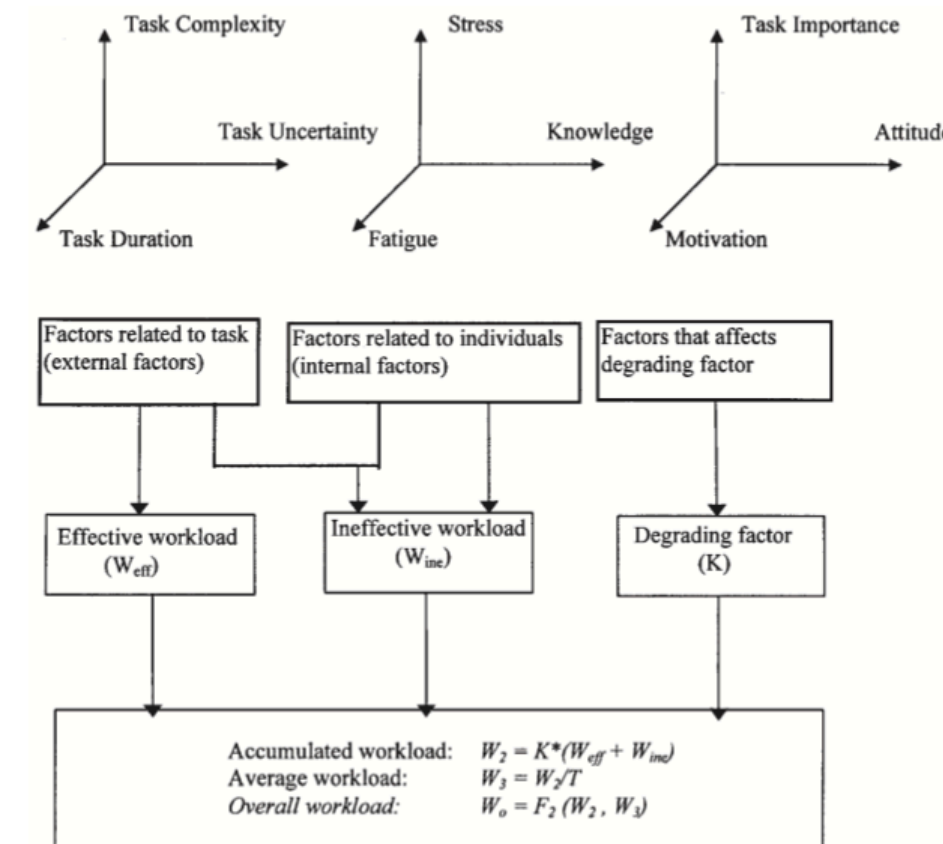


FIGURE 3 Model for predicting mental workload.

The relationship between the various workload measures is mathematically defined here:

$$\begin{aligned}
 W_p &= \text{Max}\{W_1(t)\}; \\
 W_2(t) &= \int_0^t W_1(u) du; \\
 W_3(t) &= \frac{1}{t} \cdot W_2(t); \\
 W_o(t) &= F_1[W_1(t)] = F_2[W_2(t), W_3(t)]
 \end{aligned}$$

Workload is necessarily task and person specific

- **effective workload** : workload que génère la tache naturellement -> task load
- **Ineffective workload** : tend a se réduire avec l'apprentissage et répétition de la tache

In our study, task complexity (T_c) and task type (T_t), which are task related factors, were selected to estimate the effective workload. The personal knowledge (P_k) and physical skill (P_p), which are individual related factors, were selected to estimate the ineffective workload.

If we further assume the linear correlation exists between the workload and the selected, then

$$W = W_{eff} + W_{ineff} = a \cdot T_c + b \cdot T_t + c \cdot P_k + d \cdot P_p + f$$

where a , b , c , d , and f are constants, and the parameters are defined as before.

Task complexity T_c was obtained using Zhao's (1992) approach. Task types were categorized as single task, system-paced multitask, and self-paced multitask. This classification aims to study the mental workload correlation under both single- and multitask environments. Domain knowledge of individuals (P_k), such as knowledge of HTML, were obtained using a designed questionnaire. Physical skill (P_p), such as typing and mouse movement ability, was obtained from a skill test.

State of science: mental workload in ergonomics

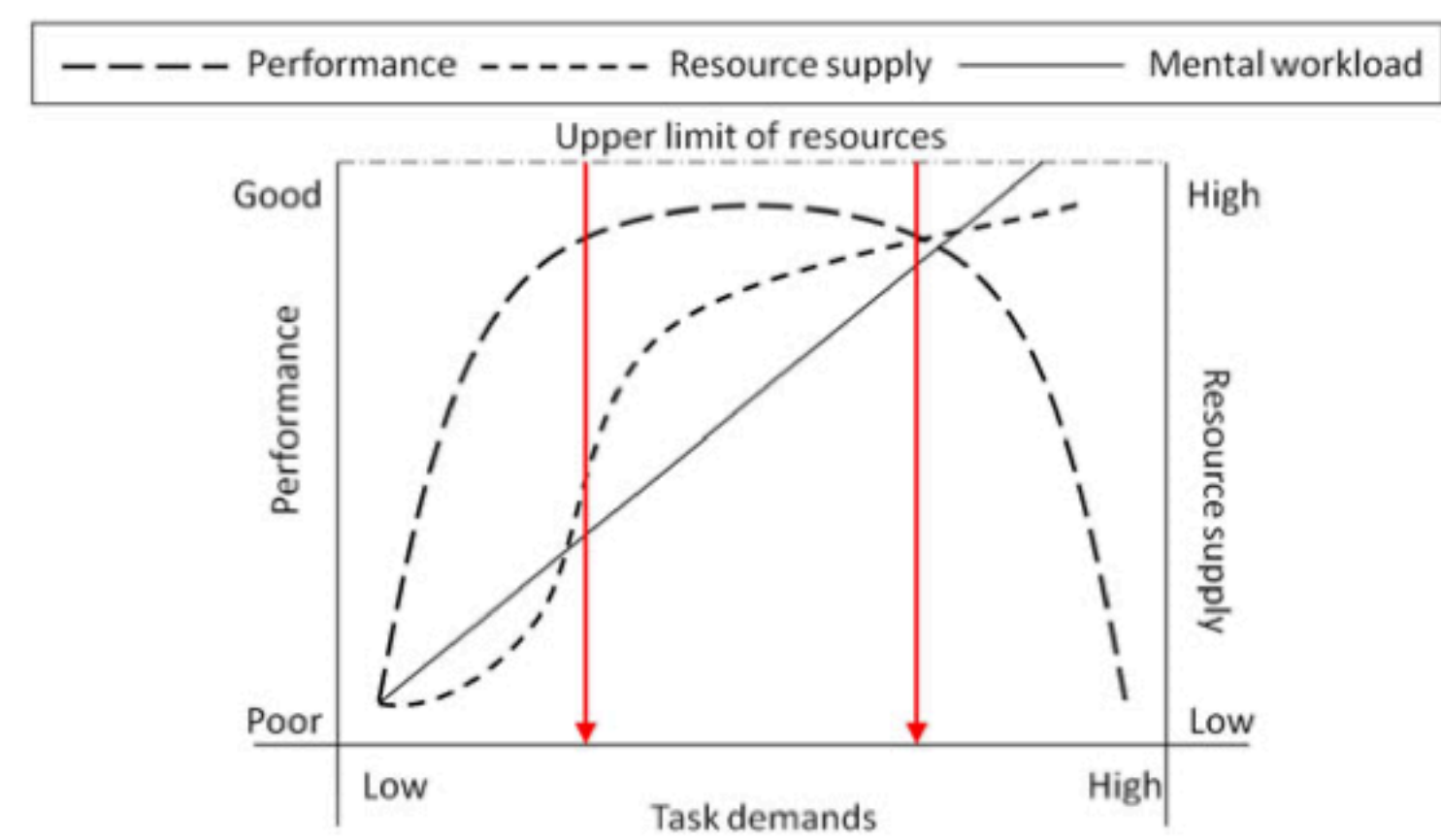


Figure 2. An interpretation of the supply–demand relationship associated with mental workload and performance, highlighting the redlines of overload and underload.

Young, Mark S., et al. "State of science: mental workload in ergonomics." *Ergonomics* 58.1 (2015): 1-17. - cité 1279

LEARNING AND FORGETTING EFFECTS

or removal between jobs, machine operations, maintenance and/or equipment replacement. According to Biskup (1999), the time needed to produce a unit decreases, usually following a negative exponential curve, as the number of repetitions of job. Hence, the actual processing time of job i if scheduled in position r is given by:

$$P_{ir} = P_i r^a \quad i, r = 1, \dots, n \tag{1}$$

where $P_{i,r}$ is the actual processing time of job i , $a < 0$ is a constant learning index, P_i is the normal processing time (i.e. the processing time without any learning effects) of job i and n is the total number of jobs. Cheng and Wang (2000) propose

3.1 Position-based learning effects

The first observation by Wright was that if the workers repeat similar tasks, then the time needed to perform a task decreases with the number of repetitions of this task. Motivated by this observation, *position-based learning model* has been proposed

3.2 Sum-of-processing time-based learning effects

To overcome the shortcomings of position-based learning effects. Kuo and Yang (2006) introduce another viewpoint of learning effects i.e. *the more time you practice, the better performance you obtain*, based on *the sum-of-actual-processing times*. In many realistic settings, in which the operating processes of a job are different or the human interactions have

after operating the job due to their experiences. Therefore, the actual processing time of a job is affected by the total actual processing time of the previous job schedules. According to Kuo and Yang (2006), the actual processing time $P_{i,r}$ is defined as:

$$P_{i,r} = \left(1 + \sum_{k=1}^{r-1} P_k\right)^a P_i$$

Azzouz, Ameni, Meriem Ennigrou, and Lamjed Ben Said. "Scheduling problems under learning effects: classification and cartography." *International Journal of Production Research* 56.4 (2018): 1642-1661. cité 199 fois

To address this issue, we developed a time-based mathematical model to quantify the impact of learning and forgetting on worker efficiency [7,50], providing a more accurate description of how efficiency changes over time.

$$p_r' = p_r \left\{ \max \left[1 + \left(\sum_{1}^{r-1} p_{[r-1]}' \right)^a, \theta \right] - 1 \right\} e^{-bt} + p_r \tag{1}$$

where p_r' represents the actual working time, p_r represents the theoretical working time, $\sum_{1}^{r-1} p_{[r-1]}'$ represents the cumulative processing time of the worker, θ is limiting parameter, $0 < \theta < 1$, a is the learning factor, $a < 0$, b is the forgetting factor, $0 < b$, and t represents the interruption or break time between two consecutive operations.

Yang W, Li S, Luo G, Li H, Wen X. A Real-Time Human–Machine–Logistics Collaborative Scheduling Method Considering Workers’ Learning and Forgetting Effects. *Applied System Innovation*. 2025; 8(2):40. <https://doi.org/10.3390/asi8020040> cité 9 fois

NO FORGETTING

$$t\{i, u, v\} = p(i, j) \times \left(v + \sum_{w=1}^{u-1} \beta_{i,w} \right)^{a_i},$$

where $a_i < 0$, $i = 1, 2, \dots, M$, $j = 1, 2, \dots, M_i$, $u = 1, 2, \dots, \pi_i$, and $v = 1, 2, \dots, \beta_{i,u}$.

TOTAL FORGETTING

$$t\{i, u, v\} = p(i, j) \times v^{a_i},$$

where $a_i \leq 0$, $i = 1, 2, \dots, M$, $j = 1, 2, \dots, M_i$, $u = 1, 2, \dots, \pi_i$, and $v = 1, 2, \dots, \beta_{i,u}$.

PARTIAL FORGETTING

A logical and reasonable range of c_i should be $a_i < c_i < 0$.

$$t\{i, u, v\} = p(i, j) \times u^{c_i} \times v^{a_i}$$

for $i = 1, 2, \dots, M$, $j = 1, 2, \dots, M_i$, $u = 1, 2, \dots, \pi_i$ and $v = 1, 2, \dots, \beta_{i,u}$.

Yang, Wen-Hua, and Suresh Chand. "Learning and forgetting effects on a group scheduling problem." *European Journal of Operational Research* 187.3 (2008): 1033-1044. cité 53 fois

proposed the general learning effect model $p_{j[r]}^A = p_j f(\sum_{k=1}^{r-1} \beta_k p_{[k]}, r)$. The sum of the normal processing times

when the learning effect reaches a given threshold. In terms of the model $p_{jr} = p_j f(\sum_{k=1}^{r-1} \beta_k p_{[k]}) g(r)$ proposed in [22], we design a general experience learning effect model as follows:

$$p_{jr} = p_j \max \left\{ (1 - \omega) f \left(\sum_{k=1}^{r-1} \beta_k p_{[k]} \right) g(r), \theta \right\} \tag{1}$$

where p_{jr} is the actual processing time of job j scheduled at position r , p_j is the normal processing time of job j , ω ($0 \leq \omega < 1$) denotes the experience factor, θ ($0 \leq \theta < 1$) is the threshold. $f : [0, +\infty) \rightarrow (0, 1]$ is a differentiable nonincreasing function so that f' is nondecreasing in $[0, +\infty)$ and $f(0) = 1$. $g : [1, +\infty) \rightarrow (0, 1]$ is a nonincreasing function with $g(1) = 1$, and β_r is the weight on position r . We assume that $0 \leq \beta_1 \leq \beta_2 \leq \dots \leq \beta_n$.

Li, Xiaoping, Yulu Jiang, and Rubén Ruiz. "Methods for scheduling problems considering experience, learning, and forgetting effects." *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 48.5 (2016): 743-754 cité 25 fois

Questionnement :

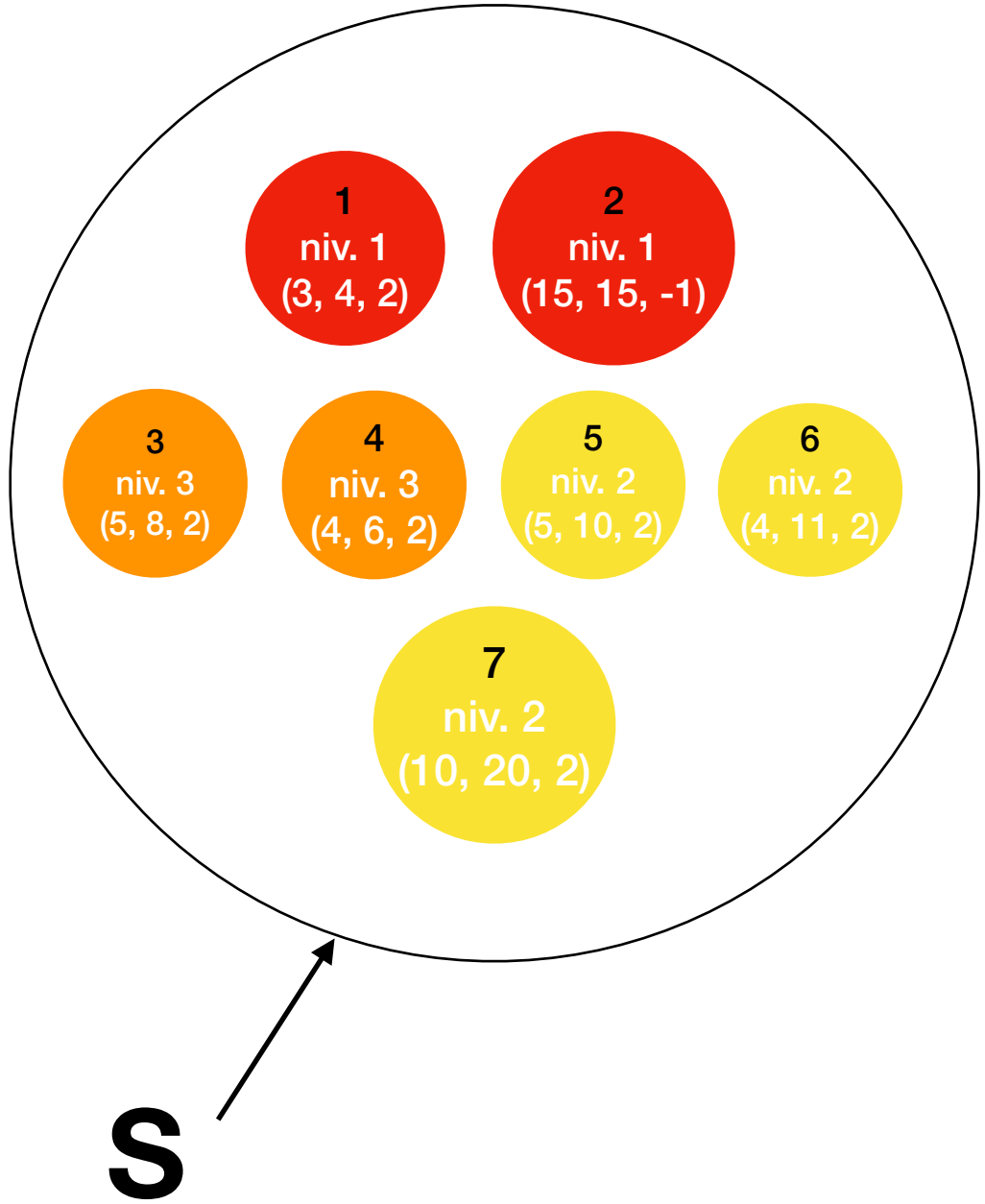
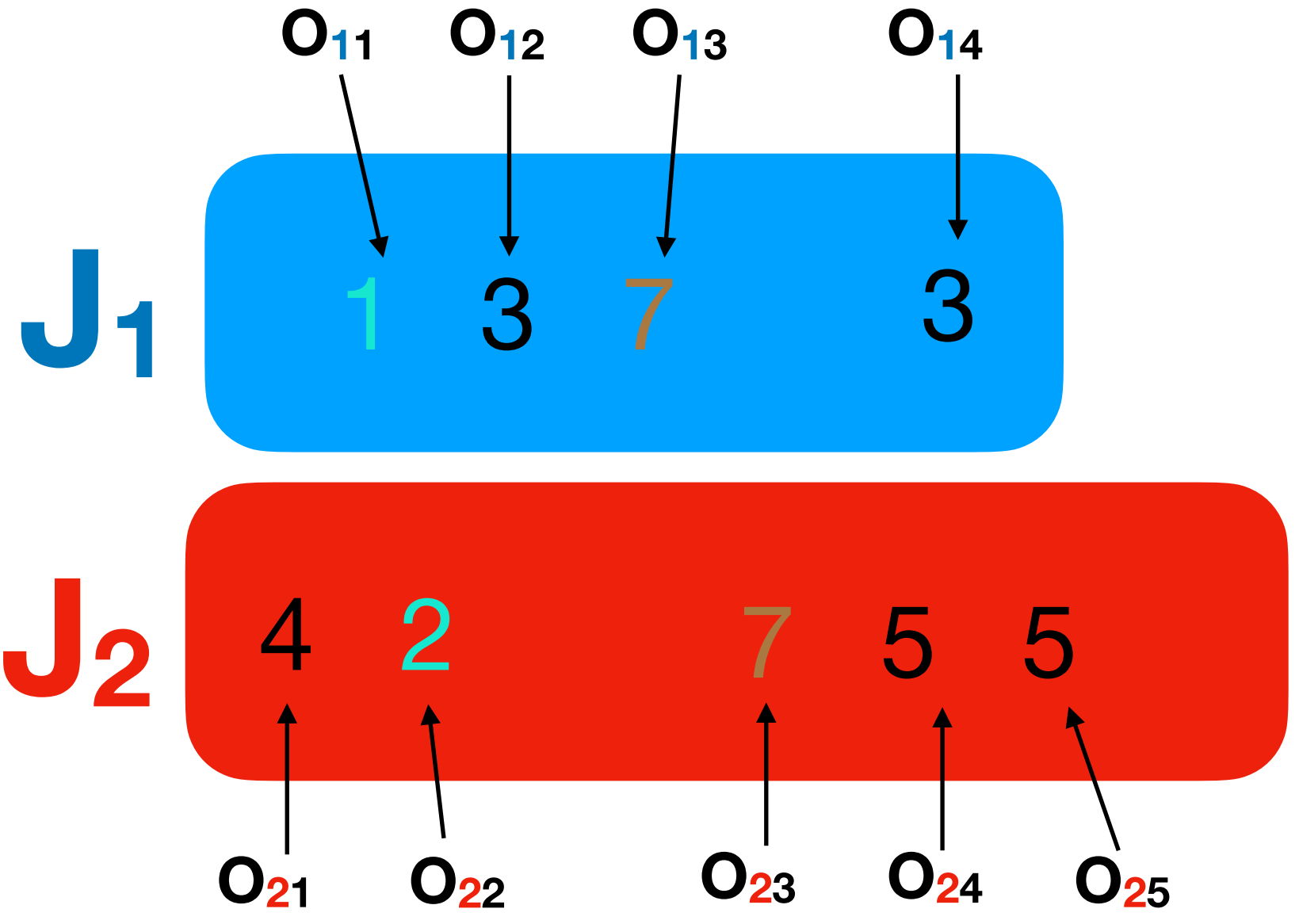
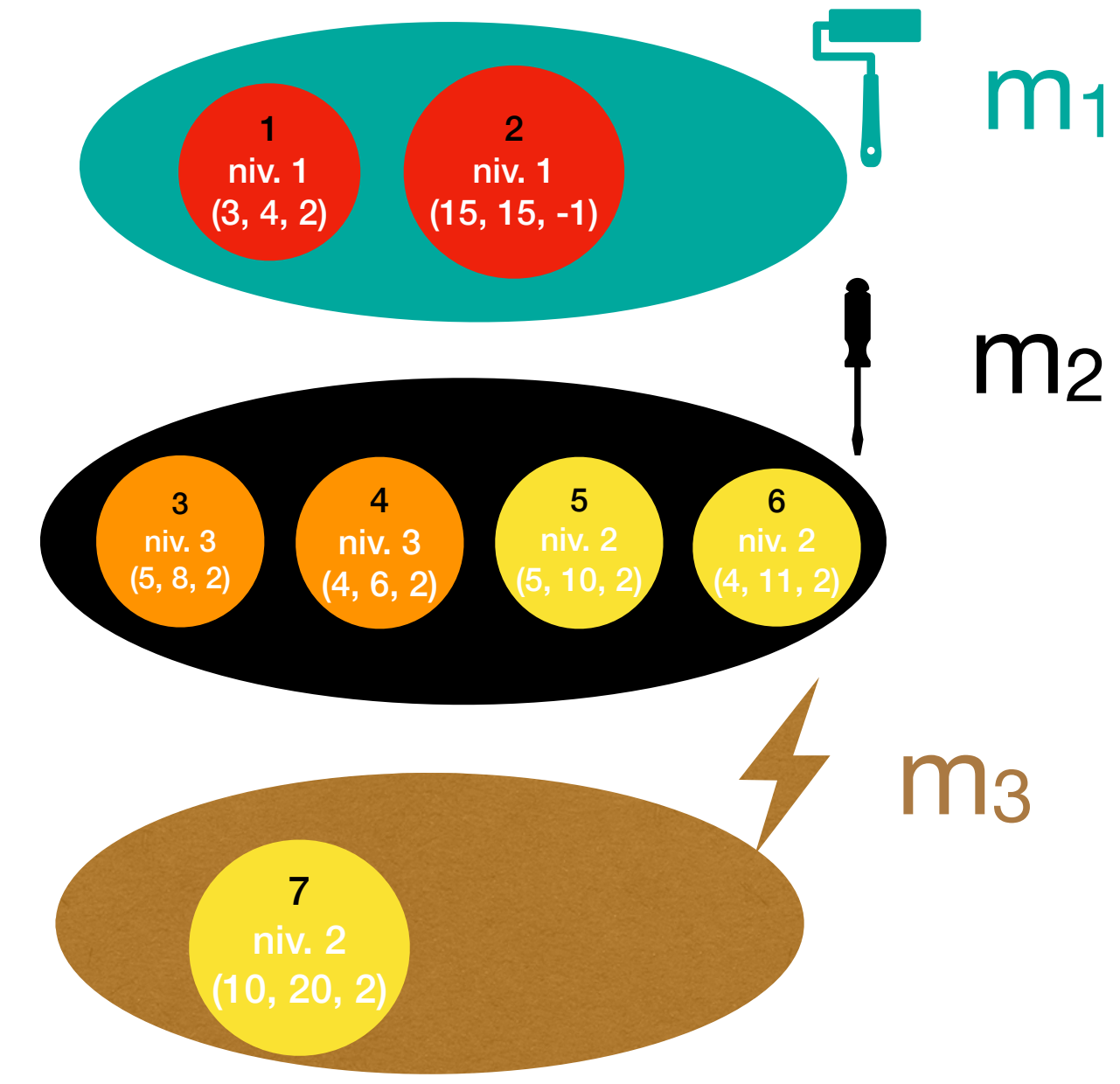
- On modélise la fatigue cognitif seulement
- Egalement la fatigue physique ?

Si on prend en compte workload mental des taches en apprentissage et collaboration, les taches fait en solo ne généreront pas de workload mental ?

Demander à l'usine le descripteur (VACP) considéré pour une tache et utilisé cette information dans le modèle pour prédire le workload mental d'une tache en apprentissage ou en collab

Structure d'une instance

Chaque opération O_{ij} appartient à l'ensemble S des taches



$L_m^k =$

w_1	1	2	1
w_2	4	3	2
w_3	1	2	1

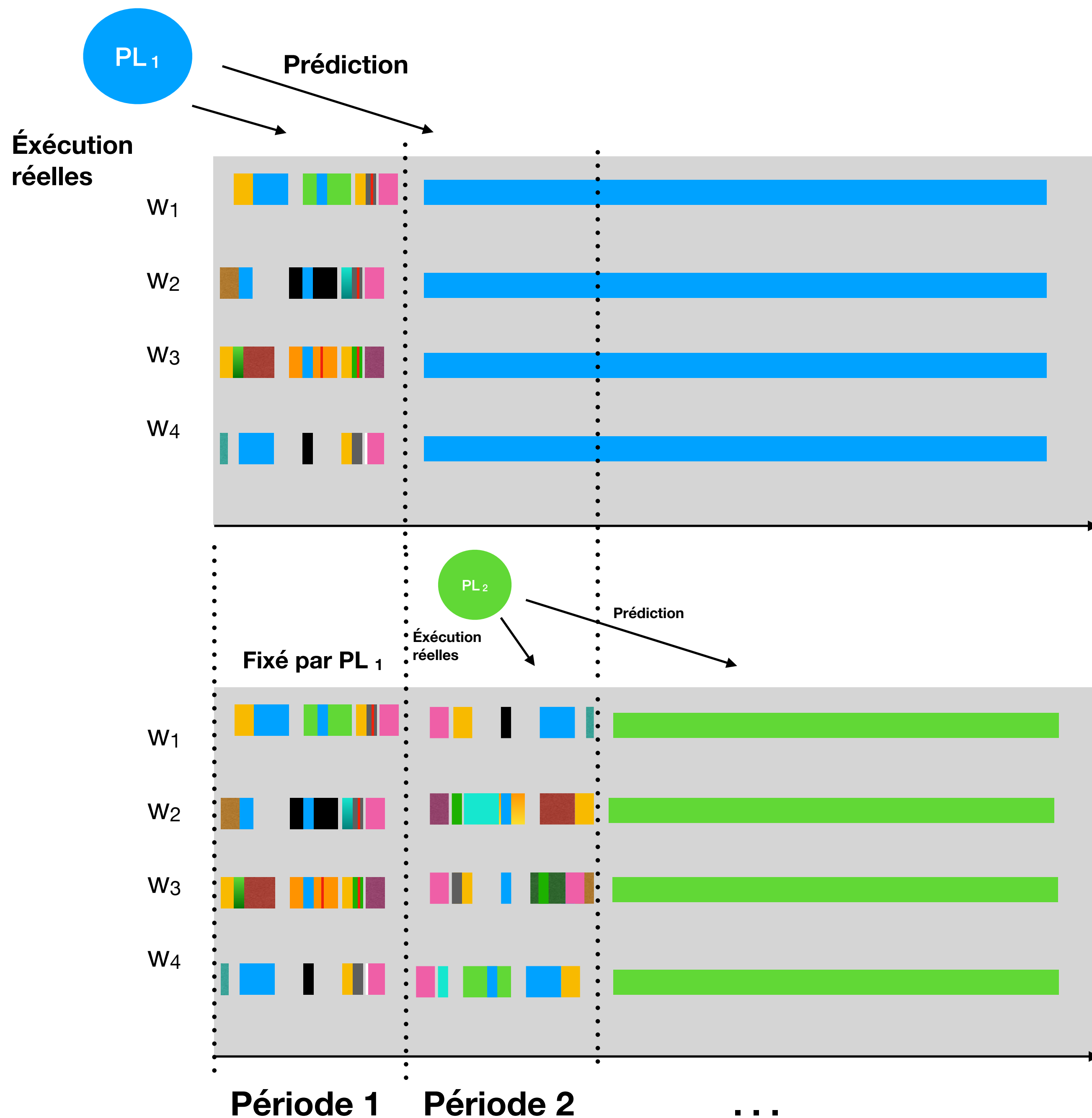
Vocabulaire:

id sub-op
Level sub-op
(x, y, z)

x: temps solo
y: temps apprentissage
z: temps collaboration
-1 : impossible

- Une tache signifie une opération O_{ij}
- $O_{ij} \in S$

Explication de quelques contraintes du modèle



Chaque PL_i doit connaître le temps de chaque période.

Pour le prendre en compte dans leurs prédiction même si obsolète car les skills peuvent évoluer

Le PL_i doit garder fixes les taches exécuté pendant la période i-1

Questionnement :

Si on considère que un PL donne l'affectation pour une semaine
-> le temps de traitement des taches doit être proportionnelle

Au moins 1 personnes Ayant un niveau de compétence requis pour l'exécution d'une tâche

Hard constraints

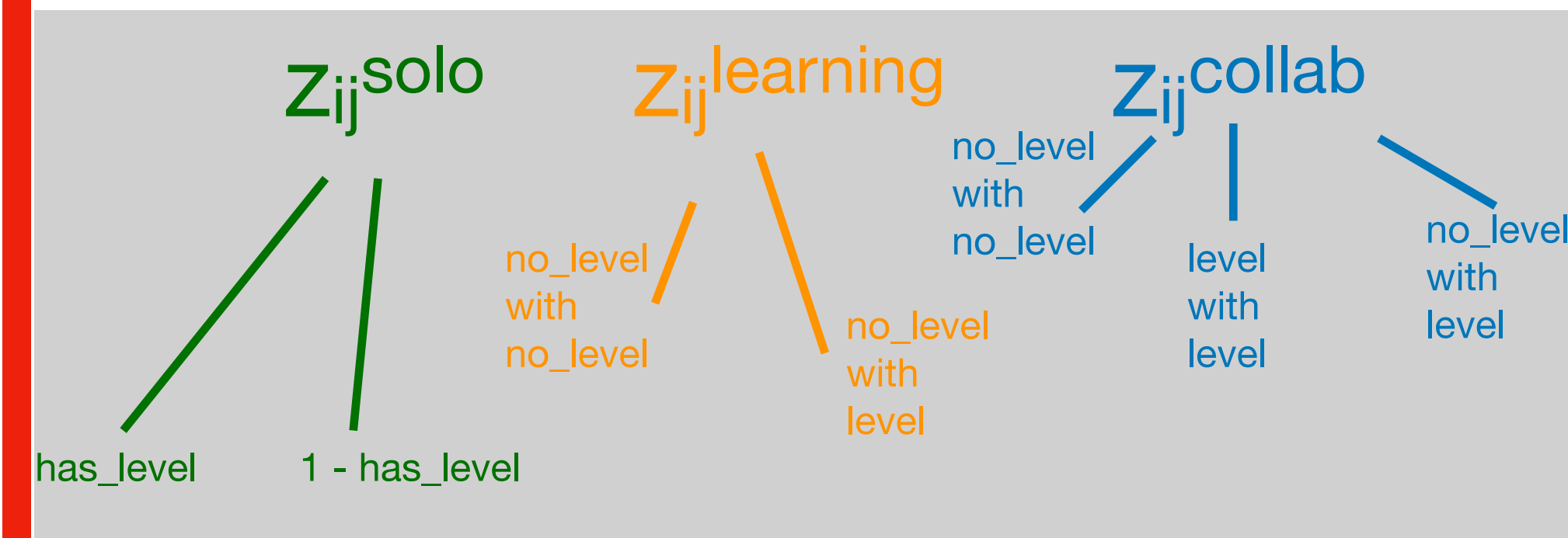
$$\forall i \in J, j \in J_i, j' \in J_i, s \in O_{ij}, k \in W, L_m^k < d_r$$

$$x_{ijsk} + \sum_{k' \in W, k' \neq k} x_{ijsk'} \geq M \cdot (x_{ijsk} - 1) + 2$$

$$\sum_{k' \in W, k' \neq k} x_{ijsk'} \cdot L_m^{k'} \geq d_r \cdot x_{ijsk}$$

Worker k n'a pas le niveau requis pour la tâche O_{ijs}

- S'il est affecté alors une autre personne doit être affecté avec lui
- Sinon contrainte non actif (big M)
- Si k fait la tâche alors la personne avec lui doit forcément avoir le niveau de compétence requis
- Sinon contrainte non actif



Soft constraints 2

Si aucun worker compétent pour O_{ij} alors on l'affecte au worker qui à la qualification la plus proche

$$x_{ijk} + \sum_{k' \in W, k' \neq k} x_{ijk'} \geq M \cdot (x_{ijk} - 1) + 2 - \text{penalty_level} \cdot L_{km}$$

$$\sum_{k' \in W, k' \neq k} x_{ijk'} \cdot L_{km} \geq D_{ij} \cdot x_{ijk} - M(\text{penalty_level})$$

Si $x_{ijk} = 1$ n'a pas le level et trouve une personne compétente :
c1: $2 \geq 2$ et c2: $3 \geq 3$ -> avec $\text{penalty_level} = 0$ et on est content

Si $x_{ijk} = 1$ n'a pas le level et personne dans l'usine possède le level requis on a :
c1: $1 \geq 2 - \text{penalty_level} \cdot L_{km}$ et c2: $0 \geq D_{ij} - M(\text{penalty_level})$

Tâche de niveau 4 et on a 3 personnes de level suivant l'ordre 1, 2 et 3

Entre : $\text{penalty_level} \cdot 1$, $\text{penalty_level} \cdot 2$, $\text{penalty_level} \cdot 3$ -> il va se caler sur le worker de niveau 3
=1, =1/2, =1/3

penalty_level : global $\in \mathbb{R}_+$
 $\text{penalty_level}_{ij}$: spécifique à chaque tâche
On souhaite **min** penalty_level

(c1): $1 = \text{penalty} \cdot L_{km}$
 $(1/L_{km}) = \text{penalty}$
 $L_{km} \leq 4$ donc $(1/4) \leq \text{penalty}$
(c2) $\Rightarrow 0 \geq 4 - M/4$
Donc **M=16**

Donc maintenant on autorise des tâches à être fait seul sans avoir la compétence

Mauvaise modél.
Pas comportement souhaité

Permettre de faire une tache si pas le niveau requis en augmentant une variable de **penalité**

SOFT CONSTRAINTS

$$\sum_{k \in W, L_{km} < D_{ij}} x_{ijk} \leq 1$$

Empeche 2 personnes non qualifiées sur la tache

$$\forall i \in J, j \in J_i$$

(55)

$$L_{km} \cdot x_{ijk} \geq D_{ij} \cdot x_{ijk} - \text{penalty_levels}_{ij} + M_4 \cdot (1 - \sum_{k \in W} x_{ijk})$$

Si k pas compétent et qu'il fait la tache alors penalty_ij vaut l'écart entre le niveau de k sur Oij et niv. De difiulté de Oij

$$\forall i \in J, j \in J_i, k \in W$$

(56)

$\text{penalty_levels}_{ij} \in \mathbb{R}_+$
 pénalité pour chaque tache fait en solo par un worker non compétent

Si k à fait tache de metier m alors pas de phénomène d'oubli. Sinon augmenter forgetting_{km}

$$\text{forgetting}_{km} \geq \text{FGT}_{km} \cdot (1 - \sum_{\substack{i \in J, \\ j \in J_i, \\ O_{ij} \in m}} x_{ijk}) + (1 - \sum_{\substack{i \in J, \\ j \in J_i, \\ O_{ij} \in m}} x_{ijk}) \cdot \text{Coef}$$

Niveau d'oubli de km Précédemment

Nombre de tache du métier m fait par k

$$\forall m \in M, k \in W$$

(57)

$\text{forgetting}_{km} \in \mathbb{R}_+$
 Niveau d'oublie du worker k pour le corps de métier m après la période considérée

- Si k a fait tache de métier m : le terme à droite de l'inégalité devient negatif
- Si k n'a pas fait de tache de métier m : alors $\geq \text{FGT}_{km} + \text{COEF}$

$$\forall i \in J, j \in J_i$$

OLD

$$\text{c1} \quad z_{ij}^{solo} + z_{ij}^{learn} + z_{ij}^{collab} = 1$$

$$\text{c2} \quad \sum_{k \in W} x_{ijk} = 1 \cdot z_{ij}^{solo} + 2 \cdot z_{ij}^{learn} + 2 \cdot z_{ij}^{collab}$$

$$\forall i \in J, j \in J_i, s \in O_{ij}, r = O_{ijs}$$

(12)

OLD

$$\text{c3} \quad LMIN_{ijs} + M \cdot z_{ijs}^1 \geq d_r$$

$$\text{c4} \quad LMIN_{ijs} - M \cdot z_{ijs}^2 \leq d_r + M \cdot \left(2 - \sum_{k \in W} x_{ijsk} \right)$$

solo : z_{ijs}^1 non constraint

appr : $z_{ijs}^1 = 1$

collab : z_{ijs}^1 non constraint

solo : z_{ijs}^2 non constraint

appr : z_{ijs}^2 non constraint

collab : $z_{ijs}^2 = 1$

Plus valable car considèrerait

- Que si level_min < Difficulté_tache alors apprentissage
- Que tache en solo implique que k a le niveau requis

$$\forall i \in J, j \in J_i$$

c1^{new}

$$z_{ij}^{solo_level} + z_{ij}^{solo_no_level} + z_{ij}^{learn} + z_{ij}^{collab} = 1$$

c2^{new}

$$\sum_{k \in W} x_{ijk} = 1 \cdot z_{ij}^{solo_level} + 1 \cdot z_{ij}^{solo_no_level} + 2 \cdot z_{ij}^{learn} + 2 \cdot z_{ij}^{collab}$$

c2.1^{new}

$$L_{km} \geq D_{ij} \cdot x_{ijk} - M_4 \cdot z_{ij}^{solo_no_level} - M_4 \cdot z_{ij}^{learn}$$

c2.2^{new}

$$L_{km} \leq (D_{ij} - \epsilon) \cdot x_{ijk} + (1 - x_{ijk}) \cdot M_4 + M_4 \cdot z_{ij}^{solo_level} + M_4 \cdot z_{ij}^{learn} + M_4 \cdot z_{ij}^{collab}$$

$$\forall i \in J, j \in J_i, k \in W, O_{ij} \in m$$

Sert si solo SANS level, mettre solo_level = 0

Sert si solo AVEC level, mettre solo_level = 1
 Si x ne fait pas , pas constraint
 Si x fait et a le niveau -> learn ou collab
 Si x fait sans niveau -> learn ou solo SANS level

$$\forall i \in J, j \in J_i$$

c3^{new}

$$level_min_{ij} + M_4 \cdot z_{ij}^{learn} + M_4 \cdot z_{ij}^{solo_no_level} \geq D_{ij}$$

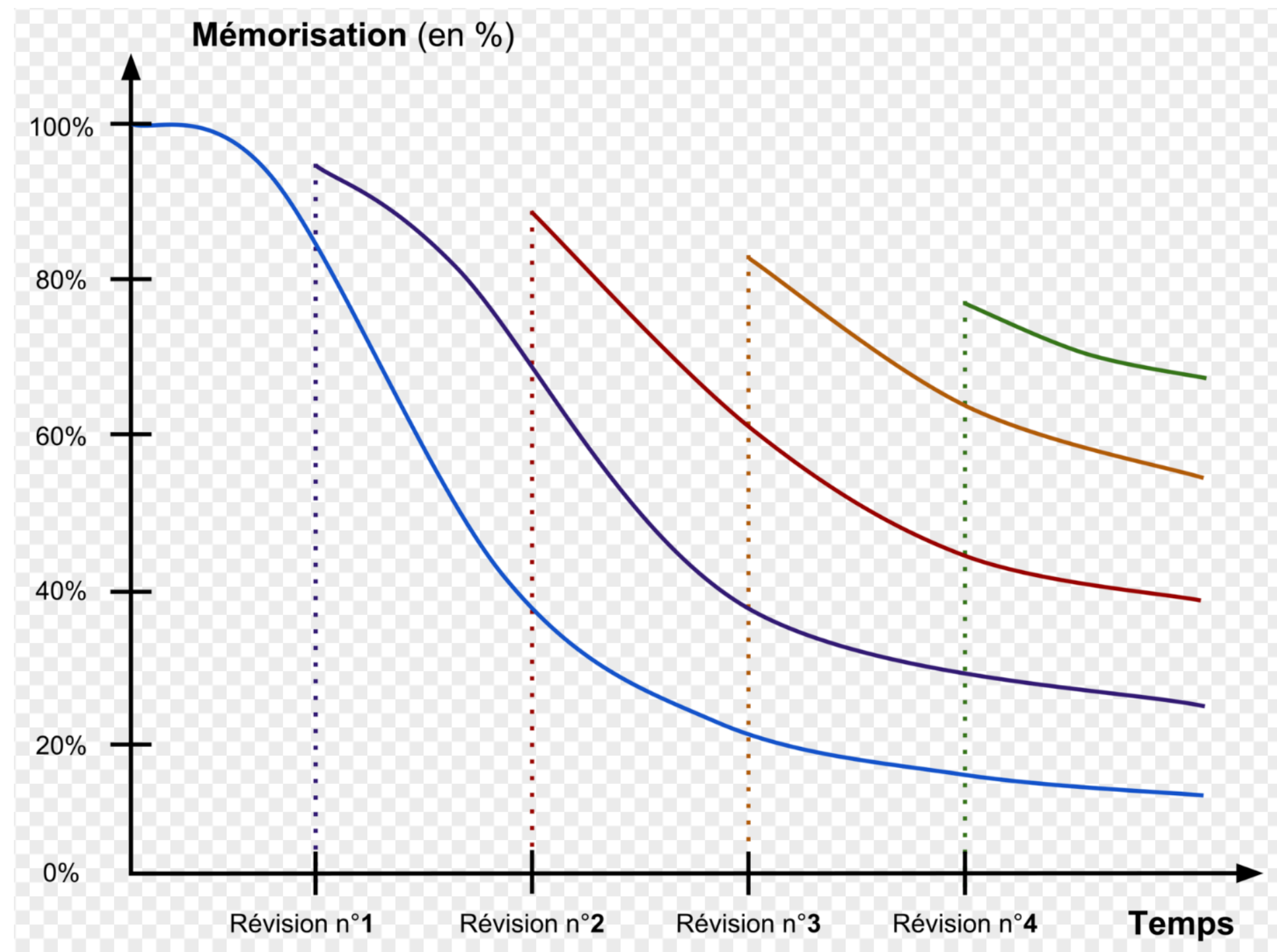
c4^{same}

$$level_min_{ij} - M \cdot z_{ij}^{collab} \leq D_{ij} - \epsilon + M \cdot \left(2 - \sum_{k \in W} x_{ijk} \right)$$

+ 20% du temps si fait en solo sans niveau

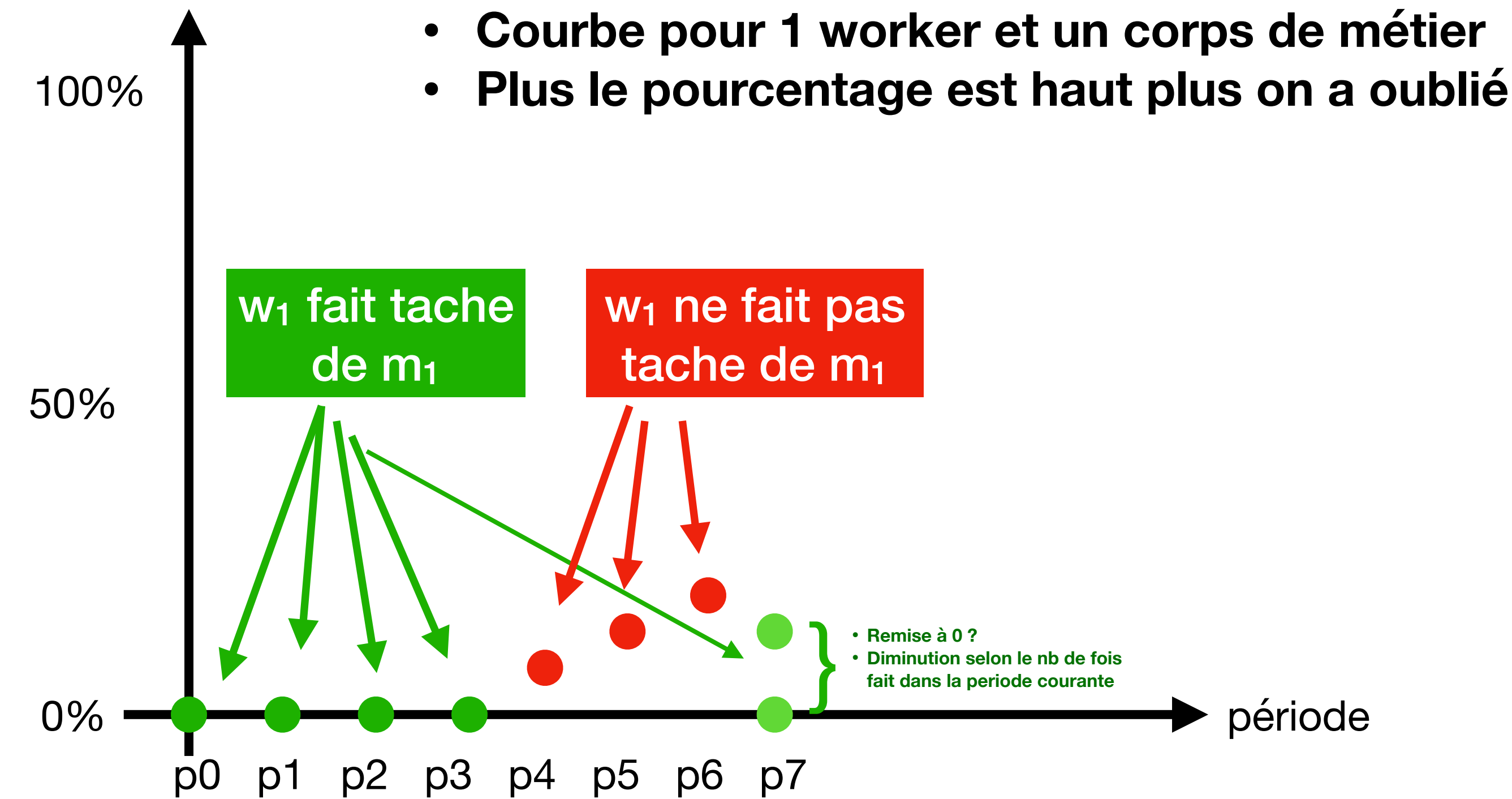
$$d_{ijk} + T_{ij}^{solo} \cdot z_{ij}^{solo_level} + (T_{ij}^{solo} \cdot z_{ij}^{solo_no_level} \cdot 1.2) + T_{ij}^{learn} \cdot z_{ij}^{learn} + T_{ij}^{collab} \cdot z_{ij}^{collab} - M(1 - x_{ijk})$$

Forgetting Level



Source : Wikipedia

Forgetting Curves



Forgetting Level

Comment agréger en corp de métier ?
Pour appliquer au levels des workers

Workers Level

	m ₁	m ₂
w ₁	3	2
w ₂	1	4

More detailed vue

1		Niv. 1	Niv. 2	Niv. 3	Niv. 4
w ₁	m ₁	-10	0	0	N/A
	m ₂	0	0	N/A	N/A
w ₂	m ₁	0	N/A	N/A	N/A
	m ₂	-10	-10	0	0

Less detailed vue

2	m ₁	m ₂
w ₁	-10	0
w ₂	0	-20

Le worker 2 après la periode considéré verrait son niveau de compétence sur les taches de métier m₂ diminuer car n'a pas fait de taches de niv. 1 et 2 de m₂

Pas très coherent

Solution rapide:

Faire diminution proportionnellement au niveau de difficulté de la tache

Le worker 2 après la periode considéré verrait son niveau de compétence sur les taches de métier m₂ ne pas diminuer car assigné à tache de m₂

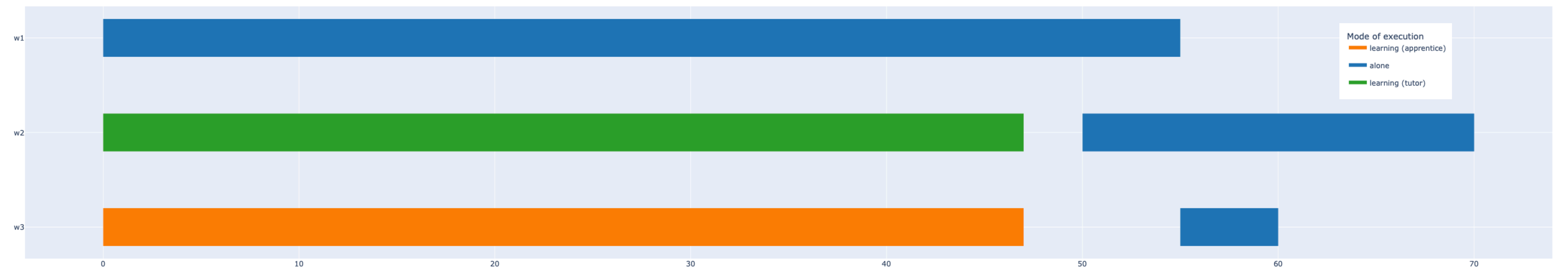
Plus coherent

Mais si fait seulement tache de niveau Niv.4 peut oublier tache de Niv. Inférieur

Alors comment s'y prendre ?

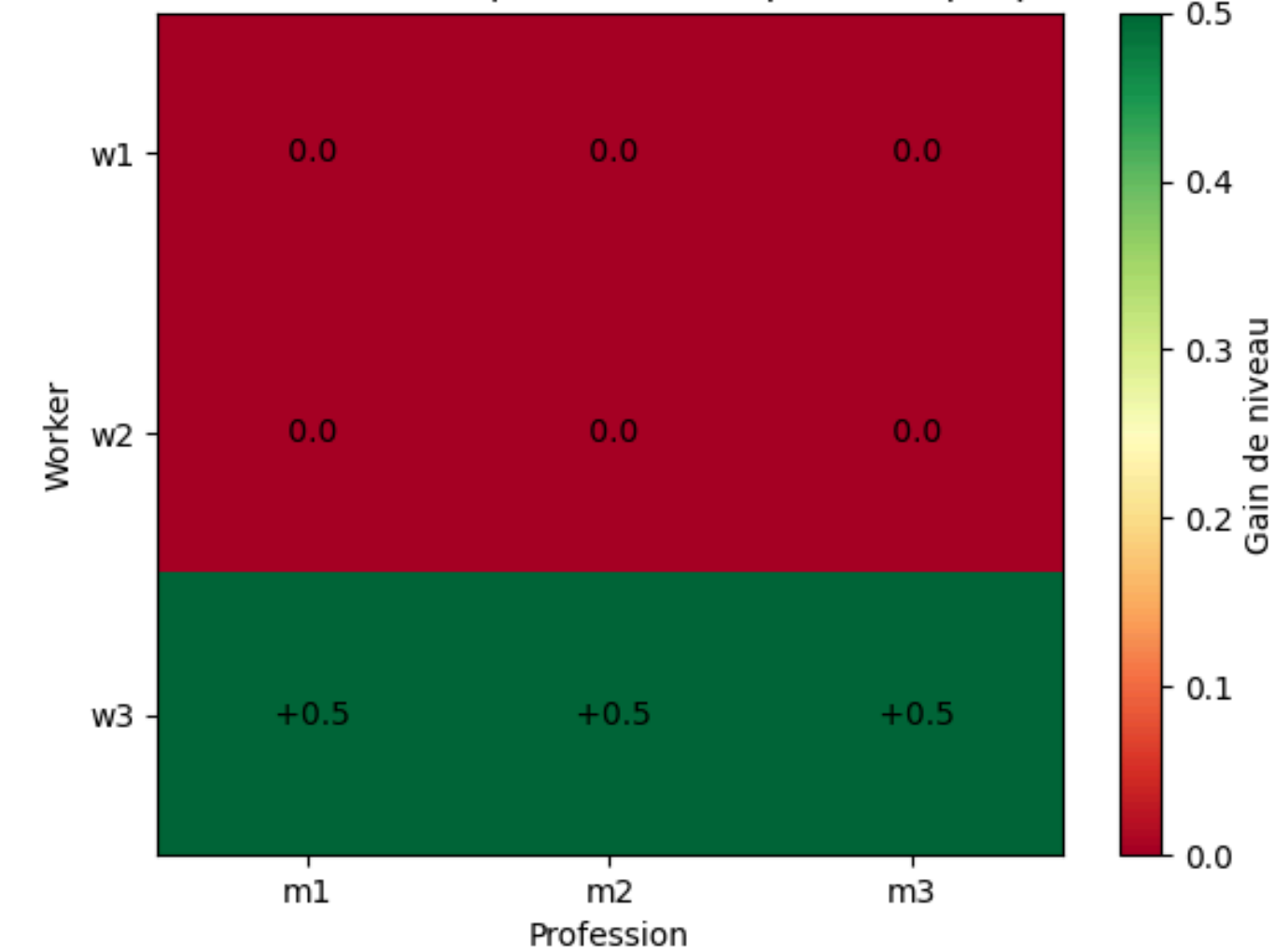
Actuellement:

- Si w à fait une tache de m alors pas d'oublie sur les taches de m
- Si n'a pas fait de tache de m, considérer forgetting effect sur ce corps de métier

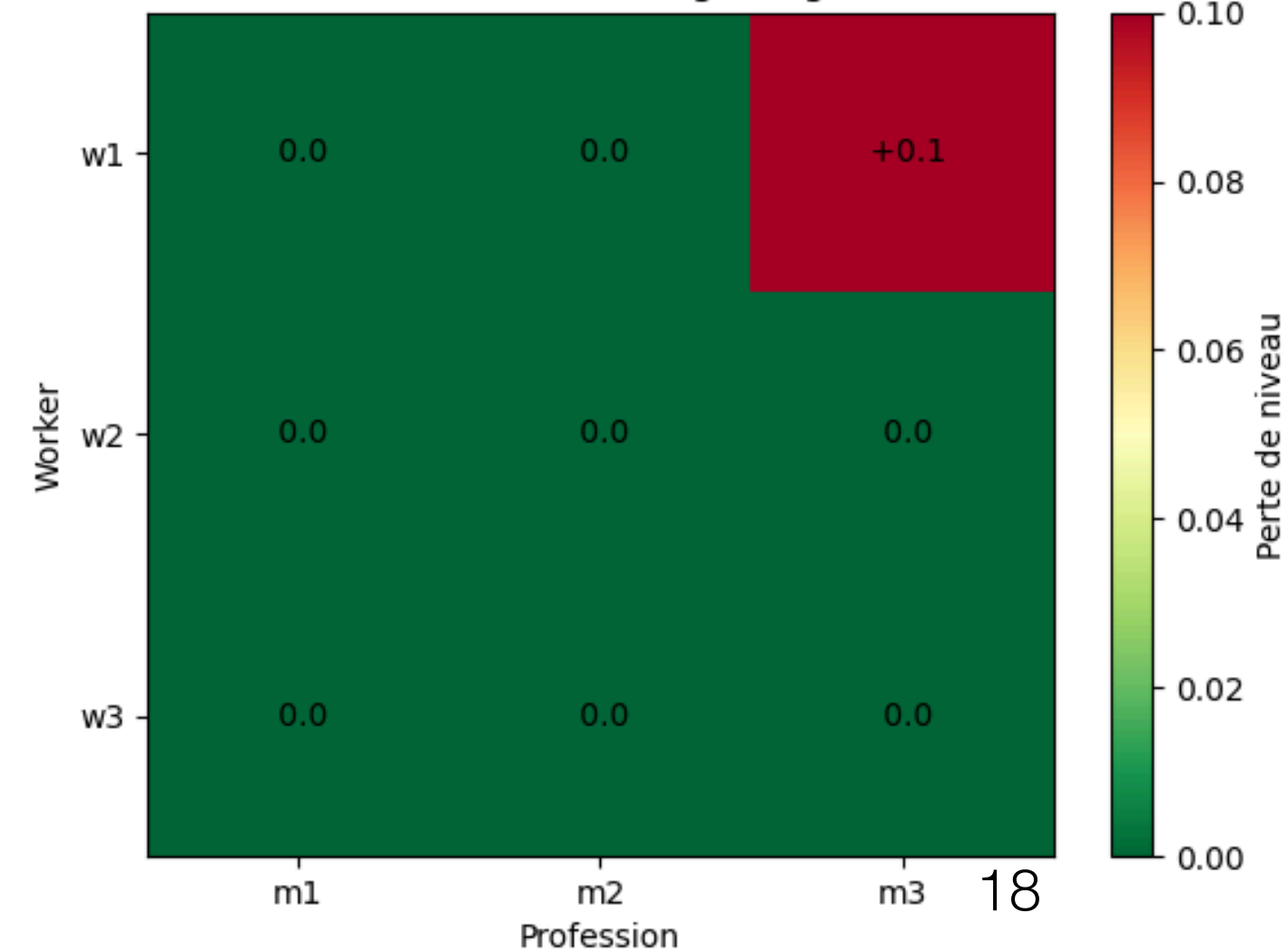


	Task	Start	Finish	Finish (var. f)	Processing time	Operation	Job	mode	Level_w	Difficulty_task	Elementary task	Metier
0	w1	0.000000	50.000000	50.000000	50.0	(1,1)	J1	alone	2.0	1	0.0	0.0
1	w1	50.000000	55.000000	55.000000	5.0	(2,4)	J2	alone	2.0	2	4.0	1.0
2	w2	0.000000	6.000000	6.000000	6.0	(2,1)	J2	learning (tutor)	3.0	3	3.0	1.0
3	w2	6.000000	27.000000	27.000000	21.0	(2,2)	J2	learning (tutor)	4.0	2	1.0	0.0
4	w2	27.000000	47.000000	47.000000	20.0	(2,3)	J2	learning (tutor)	2.0	2	6.0	2.0
5	w2	50.007001	55.007001	55.007001	5.0	(1,2)	J1	alone	3.0	3	2.0	1.0
6	w2	55.007001	65.007001	65.007001	10.0	(1,3)	J1	alone	2.0	2	6.0	2.0
7	w2	65.007001	70.007001	70.007001	5.0	(1,4)	J1	alone	3.0	3	2.0	1.0
8	w3	0.000000	6.000000	6.000000	6.0	(2,1)	J2	learning (apprentice)	2.0	3	3.0	1.0
9	w3	6.000000	27.000000	27.000000	21.0	(2,2)	J2	learning (apprentice)	1.0	2	1.0	0.0
10	w3	27.000000	47.000000	47.000000	20.0	(2,3)	J2	learning (apprentice)	1.0	2	6.0	2.0
11	w3	55.000000	60.000000	60.000000	5.0	(2,5)	J2	alone	2.0	2	4.0	1.0

Gains de niveaux de chaque travailleur pour chaque profession



Pertes de niveaux - Forgetting Effect



Si une personne fait une tache en solo sur une compétence qu'il n'a pas, faire augmenter son level ?

Lorsqu'un worker voit ses skills baisser pour un corps de métier spécifique du au Forgetting Effect:

Que se passe - t - il ? (Ex: était à 3 puis passe à 2)

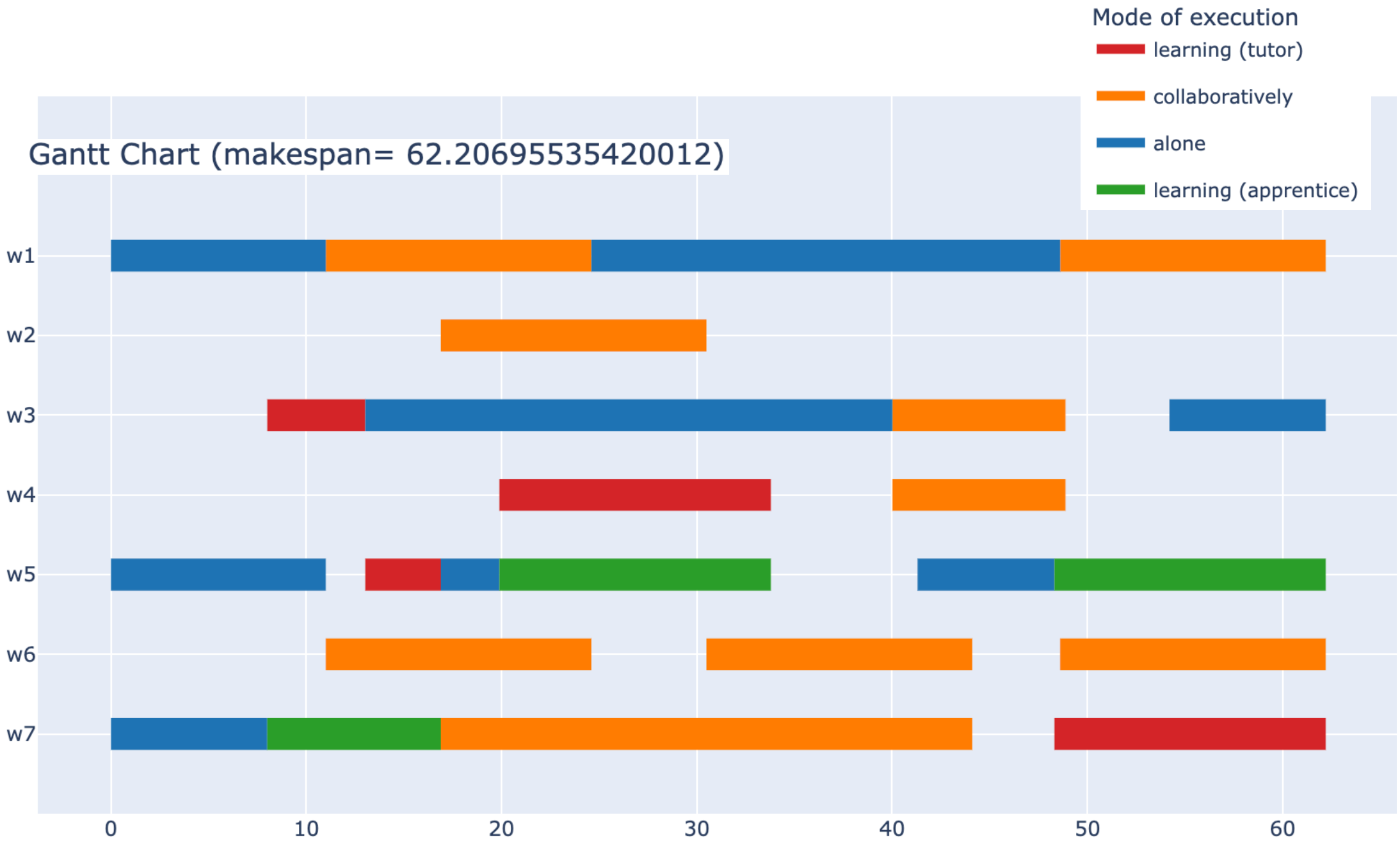
Il perd son niveau ?

Il doit réapprendre pour repasser au niveau 3 ?

Un rappel est peut etre suffisant dans la vraie vie (pas besoin de réapprendre)

Number of jobs: 4
Number of professions: 3
Number of tasks per profession: [2 2 3]
Max number of operations per Jobs: 7
Total number of tasks: 7
Total number of workers: 7
Task to profession mapping:
{0: 0, 1: 0, 2: 1, 3: 1, 4: 2, 5: 2, 6: 2}
Worker levels: shape=(7, 3)
[[2 2 2]
[1 2 1]
[2 4 4]
[3 4 3]
[4 1 4]
[3 3 2]
[1 4 2]]
Job difficulties: shape= (4,)
[2 4 2 4]
Task difficulties: shape= (7,)
[1 1 3 2 4 1 4]
Task processing times: shape= (7, 3)
[[7. 8.44322744 5.33664751]
[19. 28.32848616 13.60347768]
[11. 15.94706071 7.10176027]
[11. 13.90071907 8.86519092]
[4. 5.01818996 2.27898772]
[8. 10.04017082 -1.]
[3. 3.87381802 2.04954276]]
Jobs structure: len= 4
Job 0 : 0_(0,0) 0_(0,1) 0_(0,2) 0_(0,3) 0_(0,4) 0_(0,5)
Job 1 : 0_(1,0) 0_(1,1) 0_(1,2) 0_(1,3) 0_(1,4) 0_(1,5)
Job 2 : 0_(2,0) 0_(2,1) 0_(2,2) 0_(2,3)
Job 3 : 0_(3,0) 0_(3,1) 0_(3,2) 0_(3,3) 0_(3,4)

seed=42



Objectifs du modèle

Pour le moment:

- Max somme des skills
- Min somme des charges cognitifs

Ne donne pas des solutions équitables à priori (supposition pour la suite)

- Max Min skills
- Min Max charges cognitifs
- Ou autre ...

1. Min Makespan

- Finir les jobs rapidement

2. Max Skill

- Former des opérateurs pour qu'ils puissent avoir plus d'impact par la suite

3. Min charge cognitif des tuteurs

- Lors de la formation d'opérateur, la charge cognitive des tuteurs augmente

4. Min charge cognitive des apprentis

- Lors de la formation des apprentis, leur charge cognitive augmente

5. Min charge cognitive des tâches réalisées en collaboration

- Lorsqu'une tâche est exécutée en collaboration, la charge cognitive des opérateurs augmente

**Actuellement j'ai sommé les points 3, 4 et 5 pour ne former qu'un seul objectif.
Sinon mettre une borne sup à ne pas dépasser, pour être en bi-objectifs**